

Abdullah Naveed

✉ abdullahnaveed102@gmail.com in linkedin.com/in/abdullah-naveed-3babbb22b 🐙 github.com/Abdullahnaveed102 🌐 abdullahnaveed102.github.io

EDUCATION

- **University of Florida, Gainesville, FL, USA** Aug 2025 - Present
 - *Ph.D. in Computer Engineering*
Research: Fault tolerance and resilience for LLMs and High-Performance Computing (HPC) applications; Data compression for scientific workloads.
- **The University of Iowa, Iowa City, IA, USA** Aug 2023 - May 2025
 - *Masters in Computer Science* CGPA: 3.99/4.0
Relevant Coursework: Compiler Optimization for HPC; Distributed Systems and Algorithms; Functional Programming and Algorithm Design; Data Visualization and Technologies; Applied Machine Learning
- **Lahore University of Management Sciences, Lahore, Pakistan** Aug 2019 - May 2023
 - *Bachelor of Science in Computer Science* CGPA: 3.61/4.0
Relevant Coursework: Machine Learning; Computer Vision; Data Mining; Data Science; Network Centric Computing; Blockchain Technology

SKILLS SUMMARY

- **Programming:** Python, C++, SQL, Bash (basic), Git
- **Machine Learning / AI:** PyTorch, TensorFlow, Hugging Face Transformers, Keras, Scikit-learn, OpenCV
- **Data Science & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Jupyter
- **Systems & Tools:** LLVM, Linux, Docker, MATLAB
- **Development:** React, React Native, JavaScript, HTML/CSS, Firebase

EXPERIENCE

Argonne National Laboratory

Visiting Graduate Researcher (Remote)

Jun 2025 - Present

- Developing Transformer-based lossy compressors for Advanced Photon Source (APS) data in collaboration with researchers at Argonne. Focusing on high-throughput compression while minimizing scientific distortion.

Graduate Research Experience

- **University of Florida:** Graduate Research Assistant Aug 2025 - Present
Developing fault tolerance techniques for LLMs in safety-critical applications; designing lightweight monitoring and error detection mechanisms.
- **University of Iowa:** Research Assistant Aug 2023 - Aug 2025
Designed LLVM-based tools for instruction-level resilience analysis; evaluated compiler/runtime techniques to detect faults in HPC workloads.

Teaching Experience

- **University of Iowa:** Teaching Assistant - Database Systems Aug 2023 - Dec 2023
Mentored 30+ students on MySQL projects, debugging queries, and relational design.
- **Lahore University of Management Sciences:** Teaching Assistant - Data Structures Sep 2022 - Jan 2023
Guided 100+ students via tutorials and office hours on programming and algorithms.

PUBLICATIONS

- **BloatProfiler: Evaluating Container Debloaters:** *IEEE SecDev*, 2023. Benchmarked container debloaters for system-call reduction, CVE mitigation, correctness, and size reduction.

ACADEMIC PROJECTS

- **COVID SOP Violation Detection:** Built a real-time computer vision system with a multi-camera setup to detect masked vs. unmasked individuals and generate heatmaps for crowd density monitoring. **Python, OpenCV, PyTorch**
- **Animal Detection with Transfer Learning:** Enhanced image classification pipeline by extending a binary classifier to multi-class animal detection using pre-trained CNN models. **Python, Keras, TensorFlow**
- **Netflix Recommendation System:** Developed a personalized recommender engine using Natural Language Processing (NLP) and unsupervised learning to cluster content and predict user preferences. **Python, Scikit-learn, NLTK**
- **LUMS Mobile App:** Created a cross-platform student portal app with features for deadlines, GPA calculation, and communication. Integrated Firebase for real-time data management. **React Native, JavaScript, Firebase**

ACADEMIC SERVICE

- **Conference Sub-Reviewer:** Reviewed manuscripts for SC'24 (Supercomputing) and ISRAE, contributing to the evaluation of top-tier HPC/AI research.
- **Contributing Team Member:** Acknowledged in ProdeBench benchmarking project for contributions to implementation support and experimental evaluation.